

LJUNGBY, Sweden

WMO: 026220

Lat: 56.8525N

Lon: 13.8794E

Elev: 148

StdP: 99.56

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.0	-12.4	-18.0	0.8	-15.0	-14.6	1.1	-11.1	8.4	3.6	7.5	3.1	0.4	10	0.567

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	26.6	18.4	24.9	17.8	23.2	16.8	19.8	24.2	18.7	23.0	17.8	21.8	2.7	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	13.4	21.8	17.1	12.5	20.4	16.1	11.7	19.6	57.3	24.1	53.7	23.3	50.6	21.9	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.5	5.9	5.3	DB	-19.4	28.9	4.6	2.2	-22.7	30.5	-25.4	31.8	-27.9	33.0	-31.3	34.7	(4)
(5)			WB	-20.0	21.6	4.5	1.3	-23.2	22.5	-25.8	23.3	-28.3	24.0	-31.5	25.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	6.7	-2.0	-1.4	1.0	5.7	10.3	13.5	16.0	15.1	11.4	6.9	3.2	0.2	(6)
(7)		DBStd	7.27	4.89	4.29	4.21	3.32	3.57	2.94	2.70	2.62	2.88	3.72	3.79	4.90	(7)
(8)		HDD10.0	1784	372	319	281	135	39	4	0	1	19	107	203	304	(8)
(9)		HDD18.3	4263	630	552	538	380	250	146	83	106	209	354	453	563	(9)
(10)		CDD10.0	581	0	0	0	5	48	111	186	158	61	12	1	0	(10)
(11)		CDD18.3	18	0	0	0	0	1	3	11	4	0	0	0	0	(11)
(12)		CDH23.3	349	0	0	0	0	18	83	189	56	3	0	0	0	(12)
(13)		CDH26.7	47	0	0	0	0	1	10	31	6	0	0	0	0	(13)
(14)	Wind	WSAvg	2.4	2.8	2.7	2.6	2.5	2.4	2.2	2.0	1.9	2.1	2.3	2.7	2.7	(14)
(15)	Precipitation	PrecAvg	814	65	52	44	46	59	72	85	83	73	79	79	71	(15)
(16)		PrecMax	1041	149	134	114	93	125	151	208	140	172	166	155	165	(16)
(17)		PrecMin	585	4	0	10	10	9	0	10	14	13	11	38	20	(17)
(18)		PrecStd	116	38	30	23	22	27	34	48	36	40	36	27	33	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.7	9.3	14.5	21.1	25.1	28.1	29.6	27.1	22.8	17.1	11.7	9.6	(19)
(20)			MCWB	7.2	6.8	8.4	12.8	16.0	17.5	19.4	19.8	16.9	13.8	10.9	8.8	(20)
(21)		2%	DB	6.1	6.7	11.4	17.9	23.3	25.8	27.2	24.9	20.4	14.6	10.4	8.3	(21)
(22)			MCWB	5.5	5.6	6.9	10.6	15.2	17.4	18.9	18.4	15.8	12.9	9.7	7.6	(22)
(23)		5%	DB	5.1	5.4	8.4	15.4	21.1	23.5	25.6	23.0	18.6	13.4	9.3	7.0	(23)
(24)			MCWB	4.5	4.6	5.6	9.3	13.6	16.3	18.3	17.5	14.8	12.0	8.7	6.3	(24)
(25)		10%	DB	4.1	4.2	6.6	13.0	18.4	21.3	23.6	21.2	16.9	12.3	8.4	6.1	(25)
(26)			MCWB	3.6	3.5	4.8	8.1	12.5	15.0	17.6	16.9	13.9	11.1	7.8	5.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.3	7.5	9.3	13.1	17.9	20.3	21.4	20.9	18.1	14.4	11.2	9.4	(27)
(28)			MCDWB	7.6	8.7	13.0	19.4	23.5	25.1	26.5	25.0	21.4	15.8	11.5	9.8	(28)
(29)		2%	WB	5.7	5.9	7.4	11.5	16.4	18.1	20.1	19.4	16.5	13.3	9.8	7.8	(29)
(30)			MCDWB	6.0	6.5	10.1	16.8	21.2	23.8	24.9	23.0	19.4	14.4	10.2	8.2	(30)
(31)		5%	WB	4.6	4.6	6.1	9.9	14.7	16.9	19.1	18.3	15.1	12.3	8.8	6.5	(31)
(32)			MCDWB	4.9	5.2	8.0	13.6	18.9	22.0	23.8	21.6	17.9	13.2	9.2	6.9	(32)
(33)		10%	WB	3.6	3.5	5.0	8.6	13.2	15.7	18.2	17.4	14.2	11.3	7.9	5.5	(33)
(34)			MCDWB	4.0	4.2	6.4	12.3	17.5	20.2	22.5	20.5	16.4	12.1	8.3	6.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.0	5.7	8.4	11.6	12.0	12.2	11.6	10.8	9.9	7.5	4.7	4.5	(35)
(36)			MCDBR	4.3	6.3	11.7	16.8	16.7	17.4	16.3	13.9	12.8	8.0	4.7	4.3	(36)
(37)			MCWBR	4.1	5.0	7.9	10.3	9.3	9.7	9.0	8.4	8.6	6.2	4.4	4.1	(37)
(38)		5% WB	MCDWB	4.4	5.5	10.2	13.9	13.4	14.4	14.0	11.7	11.4	7.2	4.6	4.4	(38)
(39)			MCWBR	4.3	4.6	7.3	9.1	8.2	8.8	8.5	7.6	8.0	5.9	4.4	4.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.294	0.322	0.342	0.378	0.376	0.367	0.389	0.388	0.357	0.345	0.321	0.284	(40)	
(41)		taud	2.273	2.315	2.348	2.319	2.384	2.432	2.384	2.391	2.465	2.419	2.340	2.240	(41)	
(42)		Ebn at Noon	589	729	814	836	863	875	844	819	795	696	551	489	(42)	
(43)		Edh at Noon	51	73	91	109	109	106	109	101	82	67	48	40	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.40	1.00	2.40	3.94	5.16	5.47	5.03	4.04	2.81	1.33	0.50	0.28	(44)	
(45)		RadStd	0.07	0.17	0.24	0.53	0.50	0.41	0.63	0.38	0.34	0.22	0.09	0.03	(45)	

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47) Regional (65 neighbors)		+0.41	N/A	N/A	N/A	N/A	N/A	N/A	-159	N/A	N/A	(47)

Nomenclature: See separate page